

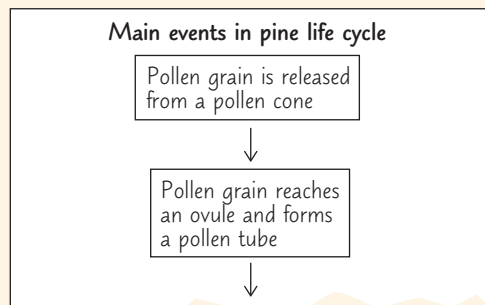
30 Plant Diversity II: The Evolution of Seed Plants

KEY CONCEPTS

- 30.1** Seeds and pollen grains are key adaptations for life on land p. 637
- 30.2** Gymnosperms bear “naked” seeds, typically on cones p. 640
- 30.3** The reproductive adaptations of angiosperms include flowers and fruits p. 644
- 30.4** Human welfare depends on seed plants p. 651

Study Tip

Make flowcharts: Make simple flowcharts showing the main events in the life cycles of a pine and flowering plant. The example shown is the beginning of a flowchart summarizing the pine life cycle.



➔ Go to Mastering Biology

For Students (in eText and Study Area)

- Get Ready for Chapter 30
- Figure 30.3 Walkthrough: From Ovule to Seed in a Gymnosperm
- Figure 30.12 Walkthrough: The Life Cycle of an Angiosperm
- Animation: Angiosperm Life Cycle

For Instructors to Assign (in Item Library)

- Scientific Skills Exercise: Using Natural Logarithms to Interpret Data
- Tutorial: Gymnosperms



Figure 30.1 In 1980, Mount St. Helens in Washington state erupted, leaving the region covered in ash and devoid of visible life. Just a few years later, seed plants such as fireweed (*Chamerion angustifolium*) had already colonized the barren landscape—an example of seed plants’ versatility and success as the dominant producers on land.



Fireweed seed

What adaptations have enabled seed plants to make up the vast majority of plant biodiversity?

Reduced male and female gametophytes

develop within parental sporophytes, protected from environmental stresses.

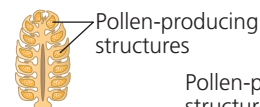
In **gymnosperms**, the gametophytes develop within cones.



In **angiosperms**, the gametophytes develop within flowers.

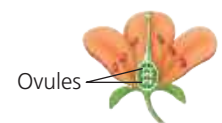
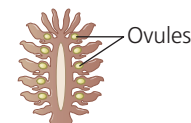


Pollen protects male gametophytes (which produce sperm) and can be transported by wind or animals.



Pollen-producing structures

Ovules protect female gametophytes (which produce eggs).



An ovule fertilized by pollen develops into a **seed**.

